

Autonomous Microstructure Alpha Discovery under False-Discovery Control

Order-Book Imbalance, Conditional Information Coefficients,
and the Adverse-Selection Barrier in Hyperliquid Perpetual Futures

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Abstract

We study the gap between statistical predictability and economic exploitability in a decentralized derivatives venue. Using 100 ms order-book and trade data for three Hyperliquid perpetual futures (BTC, ETH, SOL), we compute the Spearman information coefficient (IC) of 236 engineered microstructure features against signed and absolute forward returns at horizons from one second to fifteen minutes. Discovery and validation are performed by an autonomous platform that subjects every candidate to a five-gate replication protocol (discovery, cost, temporal, cross-symbol, and correlation gates) with Benjamini–Hochberg false-discovery-rate control, an adaptive acceptance threshold, and continuous decay monitoring—a disciplined apparatus designed to suppress the multiple-testing bias that pervades empirical asset pricing. Order-book imbalance is the dominant directional predictor, attaining IC ≈ 0.45 at the 1–5 second horizon and decaying to near zero by five minutes, with the same feature hierarchy across all three symbols and a clean directional/volatility orthogonality. The signal passes seven of eight robustness checks for BTC and six of eight for ETH and SOL. Our central result is a negative one: the *conditional* information coefficient, measured on the subset of states where a directionally-correct passive (mid-cross) fill would occur, collapses from 0.45 to ≈ 0.03 across all symbols. This is a structural adverse-selection barrier, not an estimation artifact, and it identifies execution—not signal quality—as the binding constraint. We then present a three-layer hierarchical combiner that gates fast microstructure timing on a slow directional bias, recovering positive cost-adjusted out-of-sample performance (IC of 0.18/0.25/0.36 for BTC/ETH/SOL) on a thin window. The contribution is twofold: a reproducible methodology for false-discovery-controlled microstructure research, and direct evidence that a strong, universal, robust predictor can nonetheless be economically inert under realistic passive execution.

Keywords: market microstructure, information coefficient, order-book imbalance, adverse selection, false discovery rate, cryptocurrency, perpetual futures, automated research

JEL Classification: C12, C55, G14, G17

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1 Introduction

1.1 Predictability versus profitability

A recurring failure mode in empirical finance is to mistake a statistically significant predictor for a tradeable one. A feature may correlate strongly with future returns in sample, survive out-of-sample testing, and still yield no economic profit once the mechanics of execution are accounted for. This paper provides a clean, quantified instance of that gap in a high-frequency setting, and argues that the gap—not the discovery of predictors—is where the binding constraint lies.

We study Hyperliquid perpetual futures, a decentralized derivatives exchange, at 100 ms resolution. We find that order-book imbalance predicts the direction of returns with an information coefficient (IC) of approximately 0.45 at the 1–5 second horizon—a very strong microstructure signal that is universal across BTC, ETH, and SOL and robust across days, intraday windows, and volatility regimes. Yet when we condition the IC on the event that a directionally-correct passive limit order would actually fill, the coefficient collapses to ≈ 0.03 . The predictability is real; the profitability, under a passive mid-cross fill model, is not.

1.2 The horizon mistake

The investigation began from a diagnostic failure: a library of roughly twenty-five trading algorithms, all evaluated at a 100-minute forecast horizon using five-minute bar aggregation,

were uniformly unprofitable. The cause was an aggregation artifact. The predictive content of microstructure features lives at the 1–5 second scale; aggregating to five-minute bars and forecasting one hundred minutes ahead destroys $IC \approx 0.45$ down to a weak residual ($IC < 0.07$). The features were never the problem—the evaluation horizon was. This reframing motivates a systematic, horizon-resolved IC study of the full feature set.

1.3 An autonomous discovery apparatus

Microstructure research at this scale is acutely vulnerable to data-snooping: with hundreds of features, multiple horizons, three symbols, and many candidate transformations, the number of implicit hypotheses is large, and ordinary p -values are not credible. We therefore conduct discovery through an autonomous platform whose design is itself a contribution. Every candidate signal must independently clear five replication gates—discovery (IC and its incremental component), cost, temporal replication, cross-symbol replication, and correlation deduplication—after which Benjamini–Hochberg false-discovery-rate (FDR) control is applied per cycle, the acceptance threshold adapts upward as strong signals accumulate, and registered signals are continuously monitored for decay. The platform ingests order-book and trade data in Rust at 100 ms and runs the discovery loop without human intervention. We treat it here as the *experimental apparatus*: the empirical results below are the output of a discovery process explicitly engineered to make false discoveries unlikely.

1.4 Contributions

1. A horizon-resolved information-coefficient map of 236 microstructure features across three cryptocurrency perpetuals, establishing order-book imbalance as the dominant directional predictor ($IC \approx 0.45$ at 1–5 s, half-life ≈ 15 –30 s), with eight approximately independent directional axes and a clean directional/volatility orthogonality.
2. A battery of eight robustness checks (per-day, intraday, volatility-regime, bootstrap, temporal out-of-sample) demonstrating that the signal is real, stable, and universal.
3. The central finding: a *conditional* information coefficient that isolates the adverse-selection barrier. Conditioning on directionally-correct passive fills collapses IC from 0.45 to ≈ 0.03 , identifying execution as the binding constraint.
4. A three-layer hierarchical combiner that gates fast timing on a slow directional bias and recovers positive cost-adjusted out-of-sample performance, the first architecture in this programme to confront adverse selection directly.
5. A reproducible methodology for false-discovery-controlled microstructure research—a five-gate replication protocol with FDR control, adaptive thresholds, and information-theoretic feature selection.

2 Related Work

Order-book imbalance and price prediction. The predictive content of limit-order-book imbalance is well established. Cont, Stoikov, and Talreja (2010) model the order book as a continuous-time Markov process and link imbalance to short-horizon price moves; Cont, Kukanov, and Stoikov (2014) formalize order-flow imbalance as a linear driver of price changes. Our contribution is not the existence of the imbalance signal but its precise horizon profile, cross-symbol universality, and—decisively—its behavior conditional on execution.

Adverse selection and the cost of liquidity provision. The classical microstructure literature explains why a passive provider cannot trivially harvest a directional signal. Glosten and Milgrom (1985) and Kyle (1985) show that informed order flow imposes adverse selection on liquidity providers; a quote that fills does so disproportionately when the provider is on the wrong side. Our conditional-IC result is a direct, model-free measurement of this effect: the very imbalance that predicts an up-move is the imbalance under which a passive buy fills only after the move has reversed.

Flow toxicity. Easley, López de Prado, and O’Hara (2012) introduce VPIN as a measure of order-flow toxicity in high-frequency markets. Toxicity is the aggregate counterpart of the per-fill adverse selection we isolate; we find, consistently, that imbalance carries directional information while toxicity and intensity features carry volatility (magnitude) information, with near-zero cross-loading.

Multiple testing in empirical finance. Harvey, Liu, and Zhu (2016) document rampant multiple testing in the cross-section of returns and call for higher significance hurdles; Bailey and López de Prado (2014) introduce the deflated Sharpe ratio to correct for selection bias and backtest overfitting; White (2000) provides the reality-check bootstrap. Our discovery apparatus operationalizes this agenda: Benjamini and Hochberg (1995) FDR control, an adaptive acceptance threshold, and mandatory temporal and cross-symbol replication are built into the pipeline rather than applied post hoc.

Information-theoretic dependence. Kraskov, Stögbauer, and Grassberger (2004) provide the k -nearest-neighbor mutual-information estimator that underlies our information-theoretic feature selection; Schreiber (2000) introduces transfer entropy for directed dependence. We use these as a complement to rank correlation, with a cost-aware minimum-information threshold derived below.

Cryptocurrency microstructure. Makarov and Schoar (2020) study price formation and arbitrage across crypto venues; Aït-Sahalia and Brunetti (2024) analyze order-flow toxicity in Bitcoin. We contribute a systematic, false-discovery-controlled microstructure study on a decentralized perpetual-futures venue, a market structure distinct from the centralized spot exchanges that dominate the literature.

3 Data and the Discovery Apparatus

3.1 Market data

The data are Hyperliquid perpetual futures for three symbols (BTC, ETH, SOL) sampled at 100 ms. A Rust ingestor subscribes to the Level-2 WebSocket feed and computes a 236-dimensional feature vector per symbol every 100 ms, written to columnar storage. The primary discovery window is 2026-05-19 to 2026-05-21 (3 consecutive full days, ≈ 2.17 million ticks per symbol). The robustness study (Section 6) spans 31 dates from 2026-04-19 to 2026-06-04, of which 23 are sufficiently complete.

The feature vector spans fifteen categories computed directly from the book and trade stream: raw book state, order-book imbalance, trade flow, realized and range volatility, permutation and tick entropy, exchange context (funding, open interest), trend and Hurst statistics, illiquidity (Kyle, Amihud, Roll), toxicity (VPIN, effective/realized spread), and derived composites. Of the 236 schema features, 207 are populated in this study; the remainder depend on a position-attribution pipeline outside our scope.

3.2 The autonomous platform as apparatus

Discovery is performed by an autonomous research system organized around a four-stage loop,

$$\text{Ingest (Rust)} \longrightarrow \text{Discover (agents)} \longrightarrow \text{Replicate (5 gates)} \longrightarrow \text{Deploy (paper)},$$

with a feedback path that retires decayed signals. Four research agents operate at different timescales (tick, minute-to-hour, hour-to-day, and a cross-agent orchestrator); each generates hypotheses, tests them through the replication protocol of Section 4, and registers only survivors. The design goal is not trading per se but *disciplined discovery*: the apparatus is engineered so that a feature reaching the registry has cleared independent replication and FDR control, making the empirical claims below resistant to data-snooping.

3.3 The information coefficient

Definition 3.1 (Forward return and information coefficient). Let p_t be the mid-price and h a forecast horizon. The forward return in basis points is

$$r_{t,h} := \left(\frac{p_{t+h}}{p_t} - 1 \right) \times 10^4. \quad (1)$$

The *directional* information coefficient of a feature f is the Spearman rank correlation $\text{IC} = \text{corr}_S(f_t, r_{t,h})$; the *volatility* information coefficient replaces $r_{t,h}$ by $|r_{t,h}|$.

Spearman correlation is chosen for robustness to the heavy tails of tick returns. To suppress the strong autocorrelation of 100 ms sampling, evaluation points are subsampled every 100 ticks (≈ 10 s), giving $\approx 21,700$ near-independent observations per symbol-horizon cell; significance is reported at $p < 0.01$ (two-sided), and significant cells are marked *. Horizons examined are $\{1\text{s}, 5\text{s}, 30\text{s}, 1\text{m}, 5\text{m}, 15\text{m}\}$.

4 Disciplined Discovery Methodology

4.1 The five-gate replication protocol

Every candidate signal must independently pass five gates before registration; failure at any gate routes the hypothesis to a documented “graveyard” with its rejection reason. The protocol is designed to control the false discovery rate under heavy multiple testing.

- G1 Discovery.** The rank IC must exceed an adaptive threshold and its incremental component $\text{dIC} \geq 0.05$ (the marginal IC beyond already-registered signals).
- G2 Cost.** Gross return per trade must exceed a fee-based floor (≥ 0.1 bps), so that statistically real signals too small to overcome costs are rejected.
- G3 Temporal replication.** The IC must hold on at least one additional date (a White, 2000, reality-check in spirit).
- G4 Cross-symbol replication.** The IC must hold on at least one other symbol.
- G5 Correlation deduplication.** The signal’s correlation with every registered signal must satisfy $\rho < 0.7$, enforcing breadth.

4.2 FDR control, adaptive thresholds, and decay

At the end of each discovery cycle, Benjamini–Hochberg FDR control (Benjamini and Hochberg, 1995) is applied across all tests at $q = 0.05$: order the p -values $p_{(1)} \leq \dots \leq p_{(m)}$, find $j^* = \max\{j : p_{(j)} \leq jq/m\}$, and retain the first j^* . The acceptance threshold itself adapts as the registry strengthens,

$$\text{minIC}(t) = \max\left(\text{floor}, \text{median}\{\text{IC}_i : i \in R(t)\} \times 0.8\right), \quad (2)$$

with a floor of 0.10, so that a marginal signal cannot enter a registry already populated by strong ones. Registered signals are continuously monitored and auto-retired if their rolling IC falls below half of their discovery IC for fourteen consecutive cycles. Together these mechanisms make the registry a conservative, self-pruning set rather than a catalogue of in-sample maxima.

4.3 Information-theoretic feature selection

As a nonlinear complement to rank correlation, an information-theoretic engine estimates mutual information $I(f; r)$ via the Kraskov–Stögbauer–Grassberger k -nearest-neighbor estimator, conditional mutual information $I(f; r \mid H)$, the interaction information $\text{II}(f; r; H) = I(f; r \mid H) - I(f; r)$ (positive for synergy, negative for redundancy), and a linear transfer entropy. A feature is deemed cost-viable only if its mutual information clears a fee-based minimum,

$$I_{\min} = -\frac{1}{2} \log_2 \left(1 - (\text{fee}/\sigma_r)^2\right), \quad (3)$$

the information required to overcome round-trip cost relative to return volatility σ_r . Greedy forward selection then maximizes conditional MI gain, $f_{k+1} = \arg \max_f I(f; r \mid S_k)$, stopping when the marginal gain falls below $I_{\min}$ —selecting features that are uncorrelated-by-construction in the information metric.

5 Signal Discovery

5.1 Information-coefficient horizon profile

Table 1 reports directional IC versus horizon for the leading features on BTC. Order-book imbalance (level-1 and depth-weighted) is the strongest directional predictor in the entire feature set, peaking at $\text{IC} \approx 0.45$ at 1–5 s; positive imbalance (more bids than asks) predicts an up-move within seconds, and ask depth carries the mirror-image negative IC. The half-life is ≈ 30 s: by five minutes roughly 80% of the predictive content is gone. By contrast, the spread and a slow regime-divergence feature carry IC that *grows* with horizon, capturing slow accumulation rather than microstructure—this weak residual is exactly what the original 100-minute evaluation was inadvertently measuring.

Feature	1s	5s	30s	1m	5m	15m
imbalance_qty_11	+0.447*	+0.453*	+0.257*	+0.184*	+0.089*	+0.056*
imbalance_depth_weighted	+0.427*	+0.434*	+0.255*	+0.185*	+0.081*	+0.037*
raw_ask_depth_5	−0.414*	−0.425*	−0.238*	−0.167*	−0.072*	−0.040*
flow_aggressor_ratio_5s	+0.112*	+0.112*	+0.074*	+0.051*	+0.009	−0.020
raw_spread_bps	−0.000	+0.006	+0.014	+0.023*	+0.073*	+0.139*
regime_divergence_1h	−0.020	−0.009	−0.035*	−0.043*	−0.058*	−0.074*

Table 1: Directional IC by horizon, BTC. Imbalance peaks at 1–5 s and decays; the spread and regime divergence build slowly. * : $p < 0.01$.

5.2 Cross-symbol scan and signal universality

Scanning all live features across BTC, ETH, and SOL, the order-book imbalance cluster dominates directional prediction uniformly: peak IC of 0.40–0.47 at 1–5 s on every symbol (Table 2). The decay rate is symbol-dependent—half-lives of ≈ 30 s (BTC), ≈ 20 s (ETH), and ≈ 15 s (SOL)—with the thinner book discovering price faster. The hierarchy of features is identical across symbols, indicating a structural rather than asset-specific phenomenon.

Feature	BTC	ETH	SOL	Horizon
imbalance_qty_11	+0.453	+0.447	+0.466	1–5s
imbalance_qty_15	+0.456	+0.431	+0.453	5s
imbalance_orders_15	+0.435	+0.438	+0.455	1–5s
imbalance_depth_weighted	+0.434	+0.410	+0.425	5s
raw_ask_depth_5	−0.424	−0.404	−0.415	5s
raw_bid_depth_5	+0.420	+0.405	+0.406	5s
cross_obi_mean	+0.354	+0.334	+0.331	1–5s
micro_queue_position_bid	+0.307	+0.282	+0.291	5s
flow_vwap_deviation	−0.287	−0.206	−0.188	1s
micro_obi_velocity	+0.192	+0.181	+0.208	1s
flow_aggressor_ratio_5s	+0.112	+0.109	+0.111	1–5s

Table 2: Top directional features at their 1–5 s peak, all $p < 0.01$. The imbalance cluster dominates across all three symbols.

5.3 Eight independent directional axes

Within the highly-correlated directional cluster, the scan resolves eight approximately independent axes (Table 3); these are the inputs combined in Section 8.

Axis	Representative feature	IC (1–5s)
Order-book imbalance	imbalance_qty_11	0.45
Raw depth asymmetry	raw_bid_depth_5	0.42
Cross-symbol imbalance	cross_obi_mean	0.35
Queue dynamics	micro_queue_position_bid	0.31
VWAP deviation (reversion)	flow_vwap_deviation	0.25
OBI velocity (momentum)	micro_obi_velocity	0.19
Imbalance entropy	ent_permutation_imbalance_16	0.17
Aggressor flow	flow_aggressor_ratio_5s	0.11

Table 3: Eight approximately independent directional axes and representative features.

5.4 Directional/volatility orthogonality

Direction and magnitude are measured by disjoint feature sets. Imbalance features have essentially zero volatility IC; the best volatility predictors—self-exciting intensity (**hawkes_intensity**, IC ≈ 0.35 at 5 s), realized volatility, trade intensity, and effective spread—have essentially zero directional IC. This orthogonality is consistent across all three symbols and motivates the separation of *direction*, *timing*, and *sizing* in the combiner of Section 8.

6 Robustness Validation

We subject the candidate signal—order-book imbalance at 1–5 s—to eight checks. Seven are confirmatory; the eighth (the conditional IC of Section 7) is the decisive constraint.

Stability across days, hours, and regimes. Per-day IC is stable across 23 valid dates (Table 4). The signal is present 24/7 with no dead intraday zone (a slight dip at 12–16 UTC, strongest at 04–08 UTC). Splitting on median five-minute realized volatility, IC remains above 0.34 in both regimes for all symbols, and is slightly stronger in low volatility—imbalance is more informative when the book is not being disrupted.

Horizon	BTC	ETH	SOL
1s	+0.442 $\pm$ 0.069	+0.418 $\pm$ 0.099	+0.415 $\pm$ 0.052
5s	+0.433 $\pm$ 0.061	+0.400 $\pm$ 0.098	+0.355 $\pm$ 0.071
30s	+0.260 $\pm$ 0.068	+0.224 $\pm$ 0.072	+0.182 $\pm$ 0.048

Table 4: Per-day rolling IC of `imbalance_qty_11` (mean $\pm$ standard deviation across days).

Bootstrap and temporal out-of-sample. A 1000-resample bootstrap on the discovery window gives 95% confidence intervals of width ≈ 0.02 : for `imbalance_qty_11` at 5s, BTC [0.442, 0.465], ETH [0.425, 0.450], SOL [0.400, 0.425]. The one cautionary check is temporal out-of-sample (Table 5): comparing the early window (May 19–21) to the late window (Jun 2–4), the 5s-horizon IC drops ≈ 0.15 on every symbol while the 1s-horizon IC is stable or improves (BTC 0.447 $\rightarrow$ 0.502). June dates carry wider spreads, consistent with thinner books and faster price discovery: the signal is not disappearing, its half-life is shortening.

Feature (5s)	BTC early	BTC late	ETH early	ETH late	SOL early	SOL late
<code>imbalance_qty_11</code>	+0.453	+0.286	+0.438	+0.277	+0.412	+0.240
<code>imbalance_depth_weighted</code>	+0.435	+0.317	+0.395	+0.270	+0.377	+0.242
<code>raw_ask_depth_5</code>	−0.425	−0.282	−0.395	−0.261	−0.367	−0.233

Table 5: Temporal out-of-sample IC at 5s (early $\rightarrow$ late). The 5s IC weakens while the 1s IC holds—a shortening half-life.

Validation matrix. Aggregating the checks against fixed *a priori* thresholds (Table 6), the signal passes seven of eight for BTC and six of eight for ETH/SOL. The failures are the temporal-OOS decay and, for ETH/SOL, the worst-single-day floor.

Check	BTC	ETH	SOL	Threshold
Per-day IC mean (5s)	0.43 P	0.40 P	0.35 P	> 0.30
Per-day IC std (5s)	0.06 P	0.10 P	0.07 P	< 0.10
Worst single day (5s)	0.24 P	0.04 F	0.17 F	> 0.20
Intraday (all windows)	P	P	P	all > 0.20
Low-volatility IC	0.46 P	0.43 P	0.39 P	> 0.30
High-volatility IC	0.42 P	0.39 P	0.34 P	> 0.30
Bootstrap CI lower	0.44 P	0.43 P	0.40 P	> 0.30
Temporal OOS delta	0.17 F	0.16 F	0.17 F	< 0.10
Total	7/8	6/8	6/8	

Table 6: Validation matrix. The signal is real, stable, and universal; the only systematic failure is the shortening horizon.

7 The Adverse-Selection Barrier

The seven confirmatory checks establish a strong, robust, universal predictor. The eighth check asks the economic question: can a passive liquidity provider capture it?

Definition 7.1 (Conditional information coefficient). Let $\mathcal{F}$ be the event that a directionally-correct passive limit order—a buy at the best bid when imbalance predicts up, or a sell at the best ask when it predicts down—fills within a fixed tick budget. The *conditional* information coefficient is the IC computed on the subsample $\{t : \mathcal{F} \text{ occurs}\}$.

Table 7 reports the result. The unconditional IC is 0.41–0.45. Conditioning on the *any*-fill event *raises* it (to 0.46–0.53), because states near fills carry larger absolute imbalance and hence larger returns. But conditioning on the *directionally-correct* fill—a buy that actually fills when imbalance predicts up—collapses the IC to ≈ 0.03 on every symbol.

Condition	BTC	ETH	SOL
Unconditional	+0.453	+0.438	+0.412
Any fill event	+0.526	+0.516	+0.460
Buy fill (imb > 0)	+0.032	−0.061	−0.032
Sell fill (imb < 0)	−0.047	−0.027	−0.021

Table 7: Conditional IC of `imbalance_qty_11`. The directionally-correct passive fill destroys the edge.

Proposition 7.1 (Structural adverse selection). Under a passive mid-cross fill model, a directional order-book-imbalance signal is not monetizable: the conditional IC on directionally-correct fills is statistically indistinguishable from zero.

The mechanism is structural, not statistical:

1. The signal says “buy” (imbalance > 0); a passive buy is posted at the best bid.
2. The fill requires the mid to fall to the bid—an adverse move.
3. At the instant of the fill, the “up” prediction has already been invalidated; the post-fill edge is zero.

Fill prevalence rises with book thinness: directionally-correct buy/sell fills occur on 20.8%/21.7% of BTC states, 24.1%/24.6% of ETH states, and 30.8%/31.1% of SOL states. This explains the otherwise paradoxical observation that a naive passive simulator generates *more* trades on SOL but *worse* performance—more fills means more adverse selection. Proposition 7.1 is the empirical content of Glosten–Milgrom (1985) adverse selection, measured directly at the per-fill level.

8 Hierarchical Signal Combination

If passive timing alone is non-monetizable, the natural response is to trade only where the expected move exceeds execution cost. We separate the eleven leading features into three functional layers and compose them multiplicatively.

Definition 8.1 (Three-layer composite). Let $L_1 \in \{-1, 0, +1\}$ be a slow directional bias (from regime divergence, spread, and a short trend EMA), active only when its IC-weighted z -score exceeds a band; let $L_2 \in [-1, 1]$ be a fast entry signal (from imbalance, cross-symbol imbalance, queue position, VWAP deviation, and OBI velocity), *zeroed when its sign disagrees with L_1* ; and let $L_3 \in [0, 1]$ be an inverse-volatility scale (from self-exciting intensity, trade count, and realized volatility). The composite is

$$S = L_1 \cdot |L_2| \cdot L_3, \quad S = 0.3 |L_2| L_3 \text{ when } L_1 \text{ is neutral.} \quad (4)$$

The directional gate is the key device: fast signals alone are non-monetizable (conditional IC ≈ 0.03), but fast signals *confirming* a slow bias operate at horizons where the expected move exceeds cost; inverse-volatility sizing reduces exposure precisely when adverse selection on passive orders is worst. Trained with a four-fold expanding window and a 100-bar embargo on five-minute bars, the combiner attains positive cost-adjusted out-of-sample performance (Table 8); the learned IC weights track the in-scan priors, arguing against pure walk-forward overfitting.

Metric (mean over folds)	BTC	ETH	SOL
Composite IC	+0.178	+0.248	+0.359
Directional accuracy	0.557	0.576	0.594
Sharpe (cost-adjusted)	+1.25	+1.71	+2.40

Table 8: Hierarchical combiner walk-forward out-of-sample results (BTC/ETH at a 5h horizon, SOL at 3.3h owing to less data).

Honest assessment. These results rest on a thin two-day window, so the confidence intervals are wide; the monotone improvement across folds is a caution flag for look-ahead or a trending sample; the slow layer dominates, so the marginal value of the fast layers over the slow bias alone is not yet proven; and the SOL figures (shorter horizon, less data) are an upper bound. The result is presented as a proof of concept that adverse selection can be *designed around*, not as a deployable strategy. The required next steps are data accumulation, a slow-layer-only ablation, and conditional-fill analysis on real execution logs.

9 Discussion

The results invert the usual order of difficulty. Discovering a strong, robust, universal directional predictor proved straightforward; the entire economic question reduces to whether that predictor survives adverse selection. Three implications follow.

Execution is the frontier. The conditional-IC collapse (Section 7) is a property of the *fill model*, not the signal. A passive mid-cross fill is the worst case; an execution model built from actual aggressor trades may recover part of the edge, and measuring the conditional IC under such a model—rather than the unconditional IC—is the correct objective function for microstructure alpha research. Statistical significance of a predictor is necessary but not sufficient; the relevant quantity is the IC *conditional on the states one can transact in*.

The signal is perishable. The ≈ 0.15 five-second decay from May to June, with a stable one-second IC, indicates that the tradeable horizon is contracting as books thin. Any deployment must track this and adapt per symbol, with SOL fastest (half-life ≈ 15 s).

Methodology as contribution. The empirical claims are credible precisely because they are the output of a false-discovery-controlled apparatus: five independent replication gates, FDR control, an adaptive threshold, and continuous decay monitoring. We suggest this design—automated discovery with replication and FDR built into the loop rather than applied after the fact—as a template for empirical microstructure research, which is otherwise highly exposed to the multiple-testing critique of Harvey, Liu, and Zhu (2016).

10 Conclusion

Order-book imbalance predicts the direction of Hyperliquid perpetual returns with $IC \approx 0.45$ at 1–5 seconds, universally across BTC, ETH, and SOL, surviving an eight-check robustness battery administered by an autonomous, false-discovery-controlled discovery platform. The same data show, decisively, that a passive mid-cross fill model eliminates the edge through adverse selection: the conditional IC collapses to ≈ 0.03 . A hierarchical combiner that gates fast timing on a slow directional bias recovers positive cost-adjusted out-of-sample performance on a thin window, demonstrating that adverse selection can be confronted by design. The binding constraint of microstructure α is therefore not signal quality but execution feasibility, and the right object of study is the information coefficient conditional on transactable states. Future work will replace the passive fill model with an event-driven model built from per-trade data, extend the robustness study as data accumulate, and test the combiner’s slow-layer ablation.

Reproducibility

The data are Hyperliquid Level-2 order-book and trade streams for BTC, ETH, and SOL at 100 ms, ingested in Rust to columnar storage. The discovery window is 2026-05-19 to 2026-05-21; the robustness window is 2026-04-19 to 2026-06-04 (23 valid dates). The information coefficient is the Spearman rank correlation of Definition 3.1, subsampled every 100 ticks, with significance at $p < 0.01$. Five-gate replication, Benjamini–Hochberg FDR at $q = 0.05$, and the adaptive threshold of equation (2) are applied as described in Section 4. Exact code revision (git SHA), per-run data fingerprints, and the feature manifest are recorded with each result and available on request. No trading thresholds, entry logic, or position-sizing parameters are reported; the paper concerns signal quality and its conditional behavior, not a deployable strategy.

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